ASSIGNMENT COVERSHEET

UTS: ENGINEERING & INFORMATION TECHNOLOGY

SUBJECT NUMBER & NAME	NAME OF STUDENT(s) (PRINT CLEARLY)	STUDENT ID(s)
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NAME OF TUTOR	TUTORIAL GROUP	DUE DATE
Sylvan Rudduck	4	1 September

ASSESSMENT ITEM NUMBER & TITLE

Assessment Task 2: Advanced Data Visualisation

☐ I acknowledge that if AI or another nonrecoverable source was used to generate materials for background research and self-study in producing this assignment, I have checked and verified the accuracy and integrity of the information used.
☐ I confirm that I have read, understood and followed the guidelines for assignment submission and presentation on page 2 of this cover sheet.
☐ I confirm that I have read, understood and followed the advice in the Subject Outline about assessment requirements.
☐ I understand that if this assignment is submitted after the due date it may incur a penalty for lateness unless I have previously had an extension of time approved and have attached the written confirmation of this extension.

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Statement of collaboration:

Signature of student(s) ____Warit Srichairattanakul_____ Date ___1 September 2025_____

ASSIGNMENT RECEIPT

To be completed by the student if a receipt is required

SUBJECT NUMBER & NAME	NAME OF TUTOR

SIGNATURE OF TUTOR	RECEIVED DATE

STYLE GUIDE for ASSIGNMENT SUBMISSION

Before submitting an assignment, you should refer to the policies and guidelines set out in the following:

- FEIT Student Guide
- UTS Library - referencing
- HELPS - English and academic literacy support
- UTS GSU - coursework assessment policy and procedures

Unless your Subject Coordinator has indicated otherwise in the Subject Outline, you must follow the instructions below for submission of assignments in the Faculty of Engineering and Information Technology.

Writing style

It is usually best to write your initial draft in the default settings of your software without formatting. Use the following guides in your writing.

Purpose and audience: use the correct genre and language style expected for the particular task.

Language: use 'plain English' for all technical writing. More information about this language style can be found at www.plainenglish.co.uk/free-guides.html.

Use spelling and grammar software tools to check your writing. Edit your document.

Standards: always use:

- Australian spelling standards (Macquarie Dictionary)
- SI (International System of Units) units of measurement
- ISO (International Organisation for Standardisation) for writing dates and times for international documents. For example **yyyy-mm-dd** or **hh-mm-ss**. However, for most applications it is more helpful to present the date in full as **26 August 2016**.

Graphics and tables should:

- be numbered
- have an appropriate heading and/or caption
- be fully labelled
- be correctly referenced.

Presentation

Unless otherwise instructed, all assignment submissions should be **word processed** using spell-check and grammar-check software. Work should be well **edited** before submission. Use the following default settings:

Page setup: set margins at no less than 20mm all around.

Paper: print on A4 bond, double-spaced and preferably double-sided, left justified.

Font: use the software default style to provide consistency. The recommended style includes:

- 10-12 pt font
- consistent formatting with a limited number of fonts
- lines no more than 60 characters (use wider margins or columns if you need to make lines shorter)

Header should include:

- your name and student number
- the title of the paper or task.

Footer should include the page number and current date.

Cover sheet and statement of originality: all work submitted for assessment must be the original work of the student(s) submitting the work. A standard faculty cover sheet (see over) must be attached to the front of the submission. Any collaboration between the submitting student and others must be declared on the cover sheet.

Referencing

All sources of information used in the preparation of your submission must be acknowledged using the APA system of referencing. This includes all print, video, electronic sources.

Phrases, sentences or paragraphs taken verbatim from a source must be in quotation marks and the source(s) cited using both **in-text** referencing and a **reference list**.

Plagiarism is the failure to acknowledge sources of information. You should be fully aware of the meaning of plagiarism and its consequences both to your marks, position at the university and criminal liability. The plagiarism in your assignment submissions can be assessed both in hard copy and in soft copy through software such as Turnitin.

The UTS Library and UTS HELPS (web links above) provide extensive information for students on referencing correctly to support you in avoiding plagiarism.

Executive Summary

This report analyses the Australian Open finals from 1905 to 2025. The analysis focused on where champions and runner-ups come from, who the top winners are, and how match performance has changed over time.

The geographic maps show that Australia and the United States are the dominant nations. Europe has produced many champions, and Asia has become stronger in recent decades. The treemaps show the top players, with Margaret Court, Novak Djokovic, and Serena Williams standing out. Parallel coordinates charts compare champions by seeding, win rate, and match duration. These charts show that different playing styles can still lead to success. Trend charts of match duration and win rate show that finals are now longer and closer. This means matches are more competitive.

The visualisations in Tableau helped make these findings clear. Each chart type gave a different view, maps showed global reach, treemaps showed dominance, parallel coordinates compared players, and scatterplots and bar charts showed changes over time. Labels, colours, and annotations were used to highlight key points and make the results easier to understand.

Overall, the analysis shows that the Australian Open has changed from being dominated by Australian players to being a global event with more competitive finals. The visualisations not only make the data easier to read but also tell the story behind the numbers.

Data collection and preparation process

The dataset was collected from online tennis records and covers the Australian Open finals from 1905 to 2025. The raw data did not include match duration. To fill this gap, estimates of match length in minutes were created based on the scoreline. These values were placed in the Mins column so that the dataset was complete.

The dataset was kept in Excel and included text fields, categories and numerical values. Data preparation was needed to change these raw values into a structured dataset that could be used for Tableau visualisation.

Filling the Mins column

I prompted ChatGPT to fill the Mins column by estimating match duration based on the scoreline. The method considered the total number of games played, the number of tie-break sets, and the number of sets. The estimation formula applied an average minutes-per-game rate, added extra time for tie-breaks, and included a bonus for matches extending beyond two sets. Special cases such as retirements and walkovers were handled separately to avoid distortion of results.

Handling missing Champion Seed and Runner-up Seed

Some records had missing or unseeded values. To address this, I applied a simple rule, the Champion's seed was set to be one rank higher (lower number) than the Runner-up's seed, while the Runner-up's seed was adjusted to be one rank lower (higher number) than the Champion's. This approach aligned with observed tournament patterns, where Champions generally hold stronger seeding than their opponents.

Adding the Winrate column

A Winrate column was also added to measure player dominance. It was calculated as sets won divided by total sets played in the final. This ratio made it possible to compare players fairly across matches of different lengths.

Formulas and calculations applied

Calculation	Formula	Purpose
Winrate	Winrate = Sets Won / Total Sets Played	Measures the proportion of sets won in the final, showing player dominance.
Normalised Mins	(Mins – MIN(Mins)) / (MAX(Mins) – MIN(Mins))	Scales match duration between 0 and 1 to allow fair comparison across matches of different eras.
Normalised Winrate	(Winrate – MIN(Winrate)) / (MAX(Winrate) – MIN(Winrate))	Standardises winrate values for visualisation in Parallel Coordinates.
Normalised Champion Seed	(Champion Seed – MIN(Champion Seed)) / (MAX(Champion Seed) – MIN(Champion Seed))	Allows comparison of champions' seed ranks on the same scale.

Normalised Runner-up Seed	(Runner-up Seed – MIN(Runner-up Seed)) / (MAX(Runner-up Seed) – MIN(Runner-up Seed))	Standardises runner-up seed values to highlight the competitiveness of the final.

Dataset's characteristics and data types

Column	Data Type	Characteristics
Year	Ordinal	Tournament year (1905–2025), used for time-series analysis.
Gender	Nominal	Categorical split. Men or Women.
Champion	Nominal	Name of the winning player.
Champion Nationality	Nominal	Three-letter country code for the champion (e.g., AUS, SRB).
Champion Country	Nominal	Full country name of the champion.
Champion Seed	Ordinal	Player’s seed ranking, filled where missing using tournament logic.
Runner-up	Nominal	Name of the finalist who lost.
Runner-up Nationality	Nominal	Three-letter code of the runner-up’s nationality.
Runner-up Country	Nominal	Full country name of the runner-up.
Runner-up Seed	Ordinal	Seed ranking of the runner-up, adjusted alongside Champion Seed.
Score	Nominal	String representation of set scores (e.g., 6–3, 7–6).
Mins	Ratio	Match duration in minutes; estimated from scorelines where not available.
1st-5th Won / Loss	Ratio	Number of games won/lost in each set, used for Winrate calculation.
Winrate	Ratio	The proportion of sets won out of total sets.

Visualisations produced in Tableau

Champion Countries (Map)

Figure 1 is a geographic map that shows the countries of the Australian Open champions. Circle size shows how many champions each country has. Colour shows gender with blue for men and orange for women. Short callouts were added so the key stories are clear without searching the whole map.

Australia is the clear leader with 50 men's champions and 44 women's champions. This reflects the roots of the event in Australia and the long record of strong local players. The United States follows with 26 men's and 18 women's champions. This shows steady strength across many eras. Europe has many champions, but they are spread across different countries. Switzerland and Serbia stand out for men because of Roger Federer and Novak Djokovic. Germany, Czechia, and Belarus have many women's champions through players such as Steffi Graf, Hana Mandlikova, Victoria Azarenka, and Aryna Sabalenka. Asia appears more in recent decades. Japan and China add mainly women's champions, which suggests the sport is spreading further.

The gender balance is different by country. Australia and the USA are fairly even between men and women. Serbia and Switzerland are stronger on the men's side. Belarus is stronger on the women's side. These patterns often come from one top player who lifts the total for their country. Djokovic lifts Serbia, Federer lifts Switzerland, Serena Williams lifts the USA women, and Margaret Court lifts Australia. The map makes it possible to link country totals to famous players and their eras.

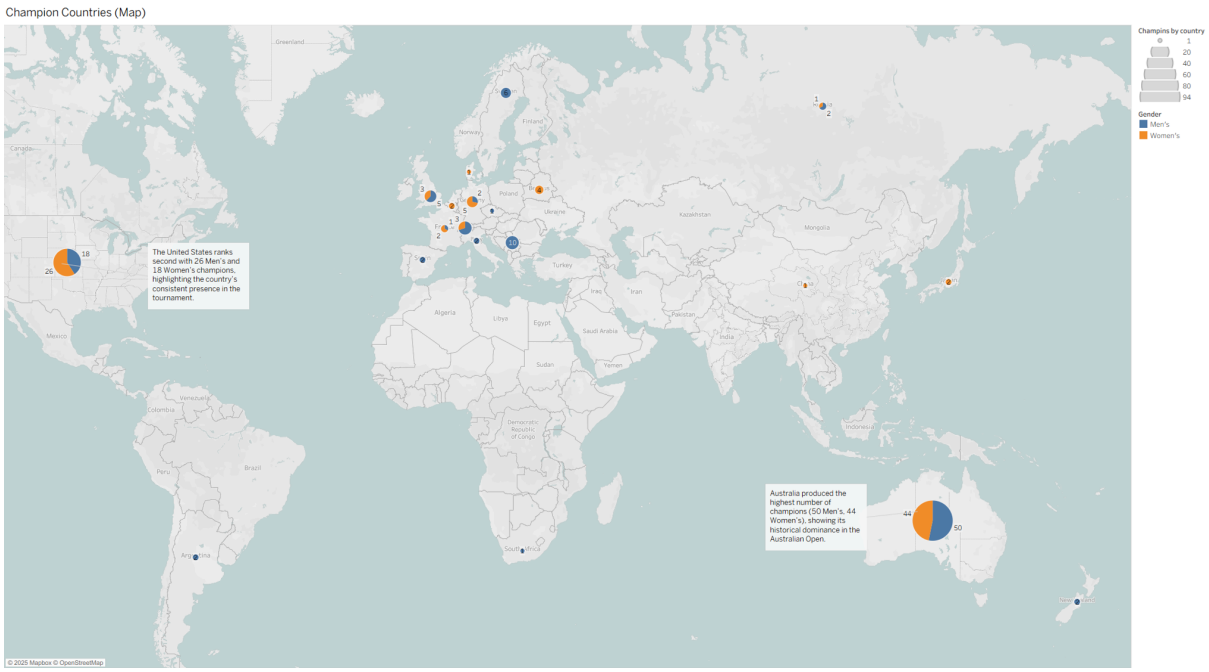

Figure 1: Champion Countries Map of Australian Open winners (1905–2025)

Runner-up Countries (Map)

Figure 2 is a geographic map that shows the countries of the Australian Open runner-ups. Circle size shows how many runner-ups each country has. Colour shows gender with blue for men and orange for women.

Australia is again the leader with 56 men's and 42 women's runner-ups. This shows not only the strength of Australian champions but also a steady record of reaching finals across both genders. The United States follows with 22 men's and 11 women's runners-up. This shows long-term competitiveness even in years without titles. Several European countries, such as Germany, Czechia, and France, also appear often as runners-up ups which shows the depth of talent spread across the continent.

The gender balance is even for Australia and the USA, but less even in other regions. Some European countries lean more towards women, which reflects the impact of strong female players in different eras. The map shows that being runner-up is also an important measure of tennis power because it means a country can reach finals often, even if it does not always win.

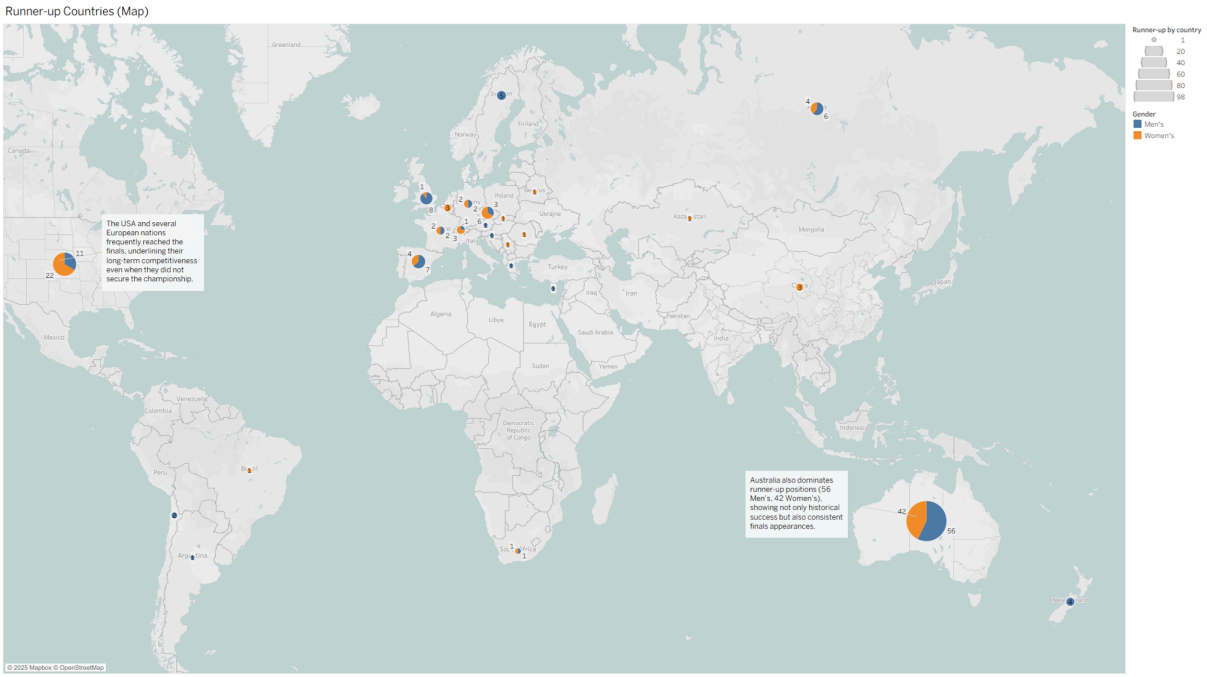

Figure 2: Runner-up Countries Map of Australian Open winners (1905–2025)

Top Winners Treemap

Figure 3 is a treemap that shows the Australian Open champions with 5 or more titles. Each block is a player. The block size shows how many titles they won. Blue blocks are men's champions, and orange blocks are women's champions.

Margaret Court is the clear leader with 11 titles, which is the most in history. Novak Djokovic follows with 10, showing his strength in the modern era. Serena Williams stands out with 7 among women's champions. Roy Emerson, Roger Federer, and Nancye Wynne Bolton each won 6. Daphne Akhurst also appears with 5.

The chart shows the balance between men's and women's dominance. Court and Serena highlight women's success while Djokovic and Federer show men's success. It also shows Australia's strong record since Court Emerson Bolton and Akhurst all came from Australia.

The treemap makes it easy to see who the top players are and how they compare across gender and eras.

Figure 3: Treemap of Top Australian Open Champions (5+ Titles)

Comparison of Top Champions' Performance

Figure 4 is a parallel coordinates chart that compares champions with 5 or more Australian Open titles. The chart uses 4 measures. These are the average champion seed, average runner-up seed, average match duration in normalised minutes, and average win rate. Each coloured line is one champion.

The chart shows clear differences across eras and playing styles. Novak Djokovic has the highest average match duration, which shows his endurance and the physical style of modern tennis. Margaret Court stands out with a high win rate, strong seeds, and long matches, which makes her the most successful champion in tournament history. Serena Williams also shows strong results with a balance of a high win rate and competitive seeding.

Roger Federer and Roy Emerson have shorter average match durations, which reflects their dominance in faster matches. Daphne Akhurst has the lowest values overall, which shows the shorter matches and less competitive structure in the early era.

This chart is useful because it shows performance patterns across several measures in one view. It makes clear that different champions succeeded through different strengths, such as endurance, seeding, or win rate.

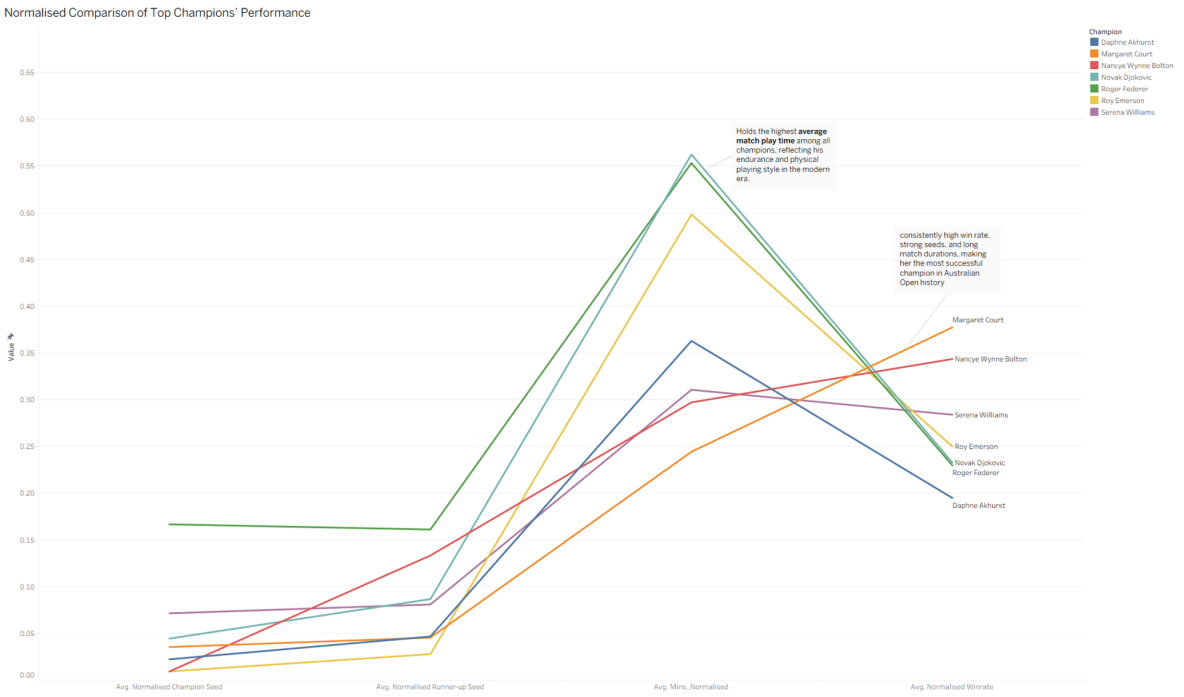

Figure 4: Normalised Comparison of Top Champions' Performance

Australian Open Champions by Country

Figure 5 is a treemap that shows the number of Australian Open champions by country. The size of each block shows how many times players from that country won.

Australia is the largest block, which reflects its strong history in the tournament. The United States is next and shows a steady record of champions across many eras. Other countries, such as Switzerland, Serbia, and Germany are smaller but still stand out because of iconic players. Roger Federer, Novak Djokovic, and Steffi Graf lifted the totals for their nations.

This chart makes it easy to see which countries produced the most champions. It highlights both the deep history of Australia and the impact of standout players from other nations. The treemap format allows quick comparison across countries at a glance.

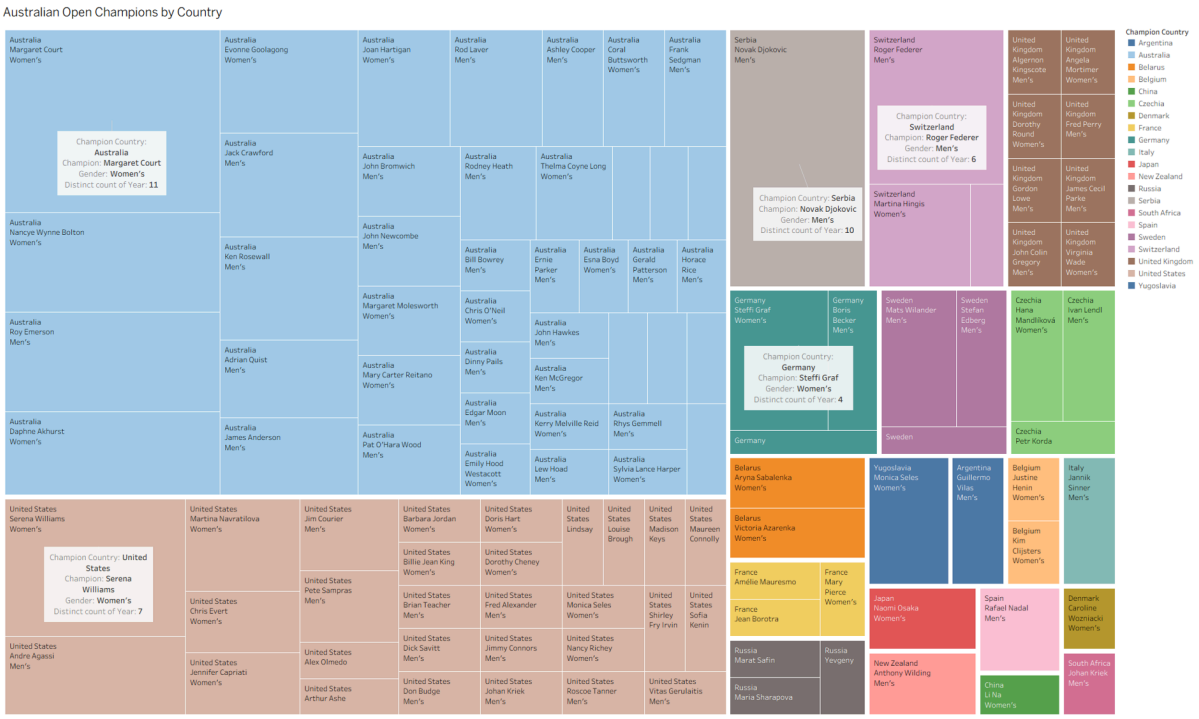

Figure 5: Australian Open Champions by Country (Treemap)

All champions' performance

Figure 6 is a parallel coordinates chart that compares Australian Open champions using four normalised measures. These are the champion seed, runner-up seed, match duration, and win rate. Each line represents one player, and the normalisation makes values easy to compare on the same scale.

The chart shows that many champions faced very strong opponents. Several lines dip at runner-up seed equal to 1, which means they defeated top-seeded players in finals. Some players, such as Amelie Mauresmo, stand out with very high win rates close to 90 percent, which shows clear dominance. Others, such as Ashleigh Barty are noted for facing consistently tough opponents.

Most players form a dense cluster in the middle of the chart. This shows that many champions had moderate seed, match durations, and winrates, which suggests balanced competitiveness across different eras. Outliers at the high end of win rate and match duration highlight the most exceptional performances in history.

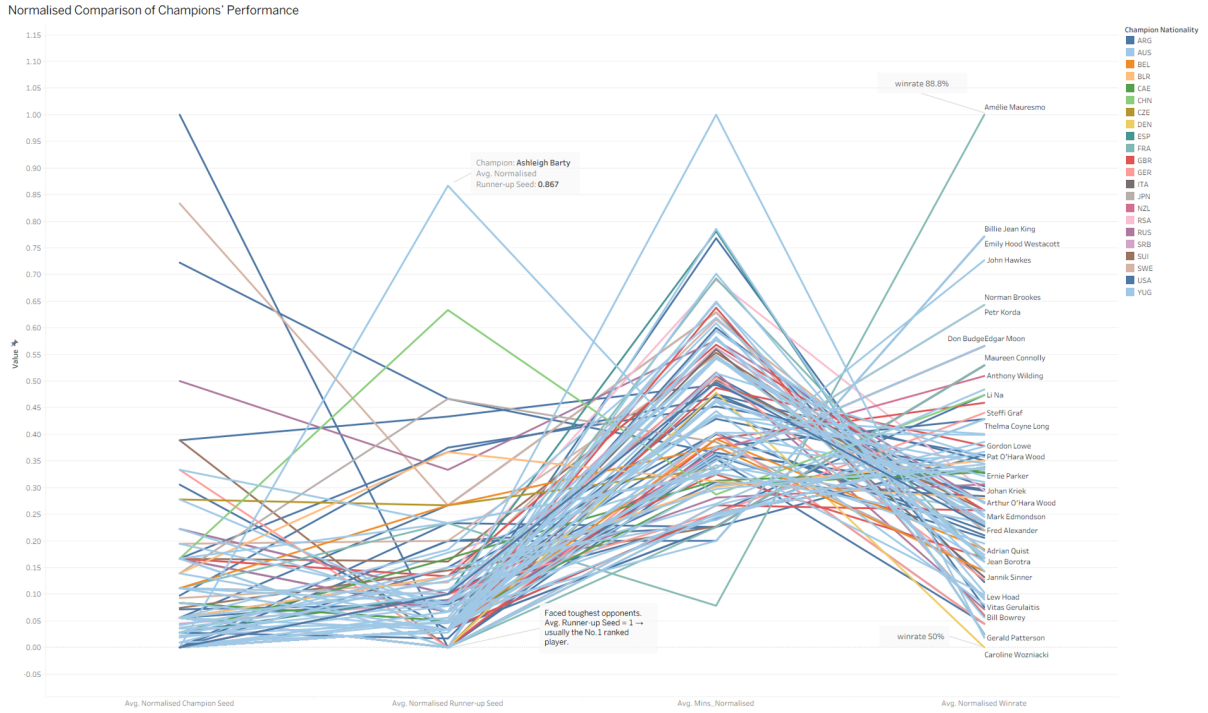

Figure 6: Normalised Comparison of Champions' Performance

Final Match Duration Trends

Figure 7 shows the trend of match duration in Australian Open finals for men's and women's singles. Each dot is one final. Blue dots are men and orange dots are women. The trend lines show the long-term change.

The chart shows that both men's and women's finals have become longer over time. The slope is steeper for men, which means their matches are more physically demanding as the years go on. Some men’s matches go above 300 minutes. These marathon matches define certain eras. Women’s finals stay closer to the 100 minute average, although in recent years, they also show signs of getting longer.

This pattern reflects the different set formats but also shows rising competitiveness in both tours. The chart makes clear that men’s finals are driving the overall increase in match duration while women’s finals are more stable.

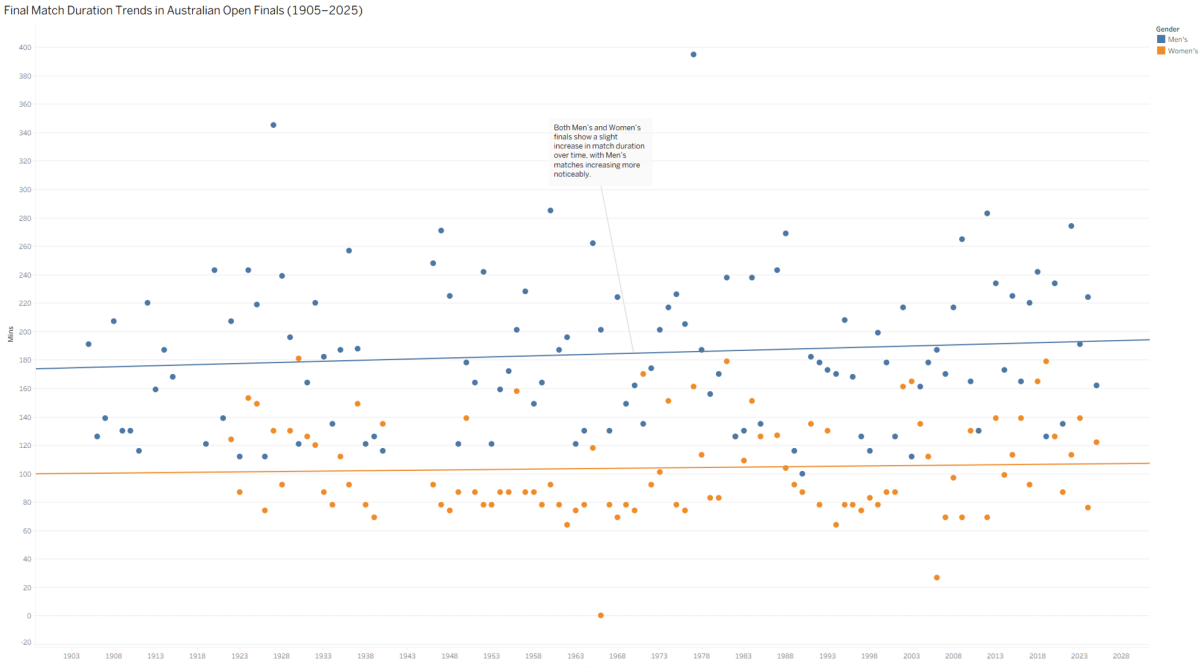

Figure 7: Final Match Duration Trends in Australian Open Finals (1905–2025)

Winrate of the Champions Trend

Figure 8 shows the average win rate of champions in Australian Open finals across more than one hundred years. Each bar is one final, and the trend line shows the overall direction.

The chart shows a gradual decline in win rates over time. In the early decades, many champions won with rates above 65 to 70 percent. This means they often dominated their opponents. In modern years, the win rates are lower and closer to 55 to 60 percent. This shows that finals are now tighter and winners must work harder against stronger opponents.

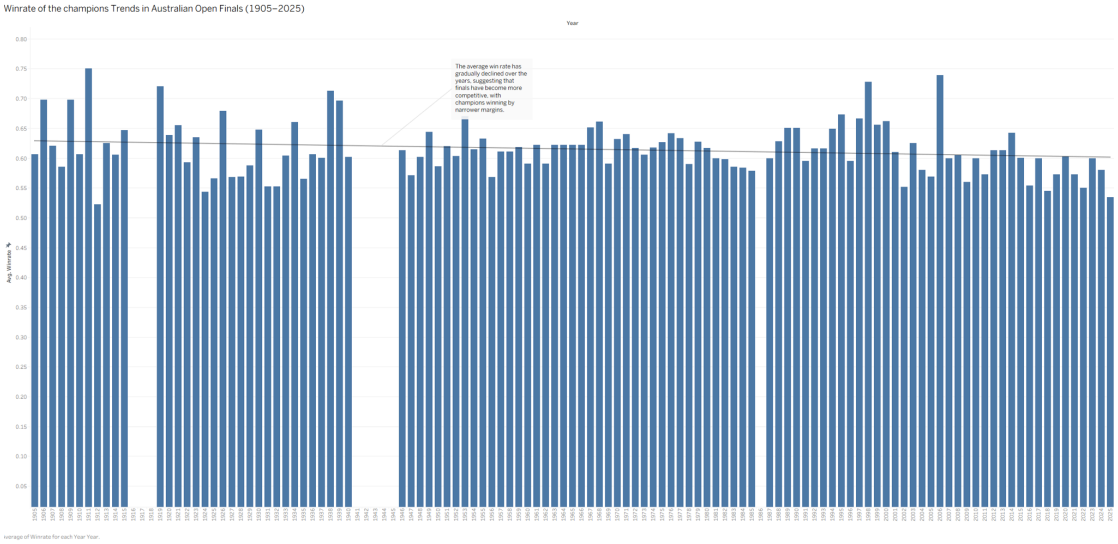

Figure 8: Winrate of the Champions in Australian Open Finals (1905–2025)

Conclusion

The analysis of the Australian Open dataset from 1905 to 2025 shows clear long-term patterns in performance, national strength, and competitiveness. The maps show that Australia is the strongest country with 50 men's and 44 women's titles, which reflects the tournament's origin. The United States is the next most consistent country. Europe contributes many champions spread across several nations. Individual stars such as Federer, Djokovic, and Graf lifted their countries almost on their own. In recent years, Asia has also appeared more often with champions from Japan and China, which shows that the event has become global.

The treemap of top champions makes clear who the biggest winners are. Margaret Court leads with 11 titles. Novak Djokovic follows with 10. Serena Williams also stands out with 7. Behind them are Roy Emerson, Roger Federer, and Nancye Wynne Bolton. This chart makes it easy to see at a glance who shaped the history of the tournament.

The parallel coordinates chart shows different strengths. Djokovic is known for endurance with long average match times. Court shows a very high win rate and strong seeds, which reflect dominance. Serena shows a balance between a high win rate and facing strong opponents. Federer and Emerson have shorter matches, which shows their efficiency. These patterns show that there are many ways to be a great champion.

The gender comparisons show more changes. Match durations are rising for both men and women, but the increase is sharper for men, which reflects the best of five format. Women's finals are more stable, but they are also becoming slightly longer and more competitive. Win rates have declined over time, which means finals are closer and the gap between winners and runner-ups is smaller.

Tableau helped bring these results together. The maps showed a global spread. The treemaps showed dominance. The parallel coordinates compared champions across measures. Scatterplots and bar charts showed long-term trends. Labels, colours, and annotations guided the reader to key points such as Djokovic's long matches Court's unmatched record, and the closer margins in modern finals.

In summary, the history of the Australian Open shows the shift from local dominance to a global and competitive event. Finals are now longer, tougher, and closer. Margaret Court, Novak Djokovic, and Serena Williams remain the standout champions, but each with a different style of success. The bigger picture is that tennis has become deeper and more balanced over time. Tableau visualisations made these insights clear and turned raw results into a story of how the tournament has changed.